

1. Prove "Perceptron learning algorithm — processes of perceptron learning algorithm — step 3. match predicted output p_y & output y (ground truth): [step 3 is only for single perceptron, not fit for neural network]":

⇒ set weighted sum function $z = w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n$

⇒ compute gradient descent of $z(w_0, w_1, \dots, w_n) = -\nabla z = - \begin{bmatrix} \frac{\partial z}{\partial w_0} \\ \frac{\partial z}{\partial w_1} \\ \vdots \\ \frac{\partial z}{\partial w_n} \end{bmatrix}$

⇒ use gradient descent method:

$$\vec{w}' = \vec{w} + \eta \cdot (-\nabla z) \longrightarrow \begin{bmatrix} w'_0 \\ w'_1 \\ \vdots \\ w'_n \end{bmatrix} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} + \eta \cdot \begin{pmatrix} -\frac{\partial z}{\partial w_0} \\ -\frac{\partial z}{\partial w_1} \\ \vdots \\ -\frac{\partial z}{\partial w_n} \end{pmatrix} \longrightarrow \begin{bmatrix} w'_0 \\ w'_1 \\ \vdots \\ w'_n \end{bmatrix} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} + \eta \cdot \begin{pmatrix} -x_0 \\ -x_1 \\ \vdots \\ -x_n \end{pmatrix} \longrightarrow \begin{pmatrix} w'_0 = w_0 - \eta \cdot x_0 \\ w'_1 = w_1 - \eta \cdot x_1 \\ \vdots \\ w'_n = w_n - \eta \cdot x_n \end{pmatrix}$$

⇒ $\begin{pmatrix} w'_0 = w_0 - \eta \cdot x_0 \\ w'_1 = w_1 - \eta \cdot x_1 \\ \vdots \\ w'_n = w_n - \eta \cdot x_n \end{pmatrix}$ is the same as

step 3. match predicted output p_y & output y (ground truth):

if "($p_y < 0$) != ($y = +1$)" = "($p_y = -1$) != ($y = +1$)" , then " add each $w[i]$ by $\eta * x[i]$ " = " $w[i] += (y * \eta * x[i])$, $y = +1$ "

if "($p_y > 0$) != ($y = -1$)" = "($p_y = +1$) != ($y = -1$)" , then " subtract each $w[i]$ by $\eta * x[i]$ " = " $w[i] += (y * \eta * x[i])$, $y = -1$ "

2. Gradient Descent Method vs. Stochastic Gradient Descent Method (SGD):

A. gradient descent method would compute mean value of the gradient descent for all individual training example before updating their weights.

B. stochastic gradient descent method would compute value of the gradient descent for a single training example before updating its weights.